

Study Guide for Test #3

Note: This study guide is meant to help you pinpoint specific topics that will be on the upcoming test. All information covered in lecture and assigned readings is fair game for the test regardless of whether it is mentioned specifically on this study guide.

Lower motor neurons

What is a lower motor neuron?

Where can you find LMNs?

Compare and contrast the different types of LMNs.

Describe the relationship between LMNs and muscle fibers.

What is a motor neuron pool?

What is a motor unit?

What are the 3 categories of motor units?

How do they differ?

How do they contribute to the size principle?

What is a neuromuscular junction?

How do they work?

Why is it important to think about innervation ratios?

What is a reflex?

Compare and contrast the 3 reflexes discussed in class.

What is a central pattern generator?

How does a CPG work?

Describe possible symptoms of LMN damage

Upper motor neurons

What is an upper motor neuron?

How are UMNs related to LMNs?

Where can you find UMNs?

Describe pathways from the brainstem to the muscles of the body.

What are the names of the pathways?

What is the point of origin for each pathway in the brainstem?

What type of movement is each pathway associated with?

Which of these pathways are involved in feedforward postural control?

Describe the pathways from cortex to the muscles of the body.

What are the names of the pathways?

What are the termination points of each pathway?

What type of movement is each pathway associated with?

Be able to draw a diagram of the hierarchy of motor control.

Which parts of the brain are at the top of the hierarchy?

What is a motor plan?

Describe how different parts of motor cortex are associated with different aspects of motor control.

Describe possible symptoms of UMN damage

How might symptoms differ based on where damage occurred?

Basal ganglia

What is the role of the BG in motor control?

Name the parts of the basal ganglia.

What is the basic progression for pathways from motor cortex through the BG?

Be able to draw each pathway that is important for motor control.

Identify excitatory/inhibitory connections and tonic/transient connections.

What role does dopamine play in each pathway?

How does each pathway influence different aspects of motor control?

What is disinhibition?

Why is it important in basal ganglia pathways?

Why is the BG often brought up when discussing Parkinson's disease?

How does neurodegeneration appear to be related to motor symptoms?

Cerebellum

What is the role of the cerebellum in motor control?

Name the parts of the cerebellum.

Describe how different functional subdivisions of cerebellar cortex relate to specific aspects of motor control.

Describe the roles of the different peduncles. Where information is coming

from/where information is going to.

Be able to draw pathways related to the cerebellum.

Describe the different types of cells and fibers in the cerebellum.

Be able to draw the circuits of the cerebellum.

Identify excitatory/inhibitory connections.

How might these circuits be important for modulating movements in response to motor error?

Describe possible symptoms of cerebellar damage.